



PROJECT REPORT · 2023

Coffee Sales Dashboard

A complete sales analytics study of a NYC coffee
retailer across three locations · Jan – Jun 2023

Project Overview

This report analyzes the **Coffee Sales Dashboard** dataset, capturing every transaction recorded across three New York City coffee shop locations between January 1, 2023 and June 30, 2023. The goal is to surface meaningful patterns in revenue, customer behavior, product mix, and operating hours that can guide merchandising, staffing, and marketing decisions.

Headline Metrics

\$698.8K TOTAL REVENUE	149,116 ORDERS	214,470 ITEMS SOLD	\$4.69 AVG ORDER
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Dataset Summary

Field	Value
Records (transactions)	149,116
Date range	2023-01-01 → 2023-06-30
Stores	Astoria · Hell's Kitchen · Lower Manhattan
Product categories	9 (Coffee, Tea, Bakery, Drinking Chocolate, Coffee beans, Branded, Loose Tea, Flavours, Pack
Sizes	Small · Regular · Large · Not Defined
Granularity	Per-transaction line items with date, time, store, product, qty & price

Approach & Methodology

Source Data

The workbook contains three sheets — **Transactions** (raw line-item data, 149,116 rows × 18 columns), **PIVOT** (intermediate aggregations) and **Dashboard** (the visual layer). Each transaction row records the order id, date and time, store id and location, product hierarchy (category → type → detail → size), quantity, unit price and a derived total bill.

Derived Fields

Several columns enrich the raw register data to enable time-series analysis: **Total Bill** = transaction_qty × unit_price, plus calendar dimensions **Month**, **Month Name**, **Day Name**, **Day of Week** and **Hour**. These were used as primary slicers throughout the dashboard.

Analytical Pillars

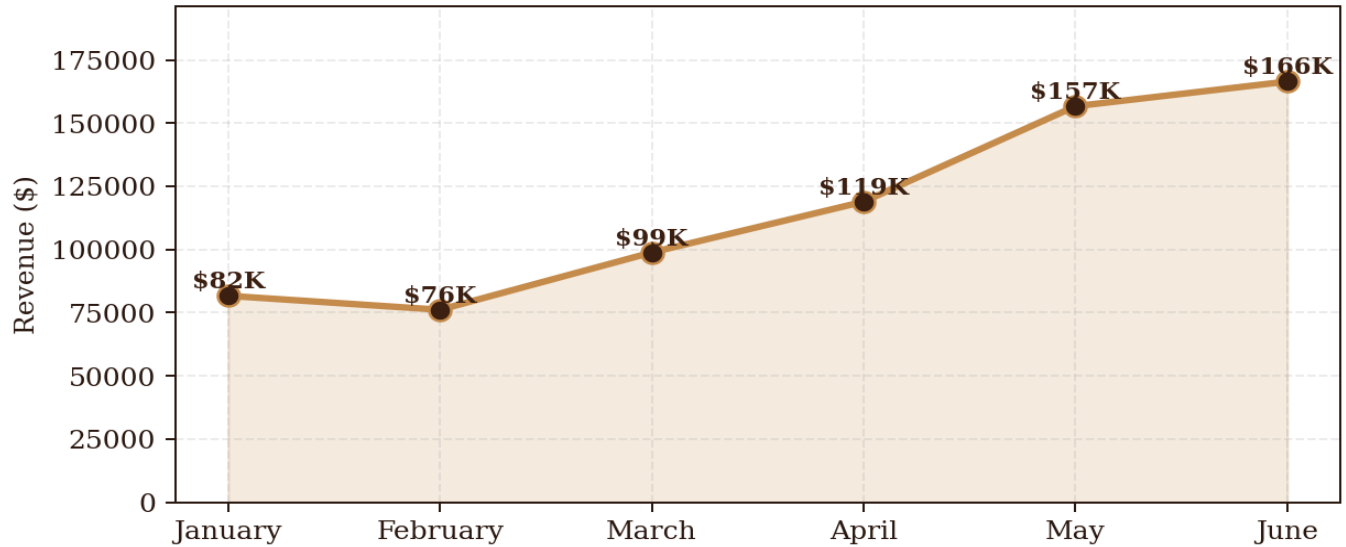
1	Revenue & Volume	Aggregate sales, order count, units sold, AOV.
2	Time-of-day / Weekday	Identify peak hours and weekday patterns to drive staffing.
3	Product Mix	Category, type, size and top-seller decomposition.
4	Store Comparison	Side-by-side performance across the three locations.
5	Trend	Month-over-month growth from Jan → Jun 2023.

Tools

Excel was used for initial pivot tables and the visual dashboard. For this report, the raw transaction sheet was re-processed in Python (pandas) to regenerate every aggregate from first principles, and matplotlib was used to render publication-quality charts.

Sales Trend Over Time

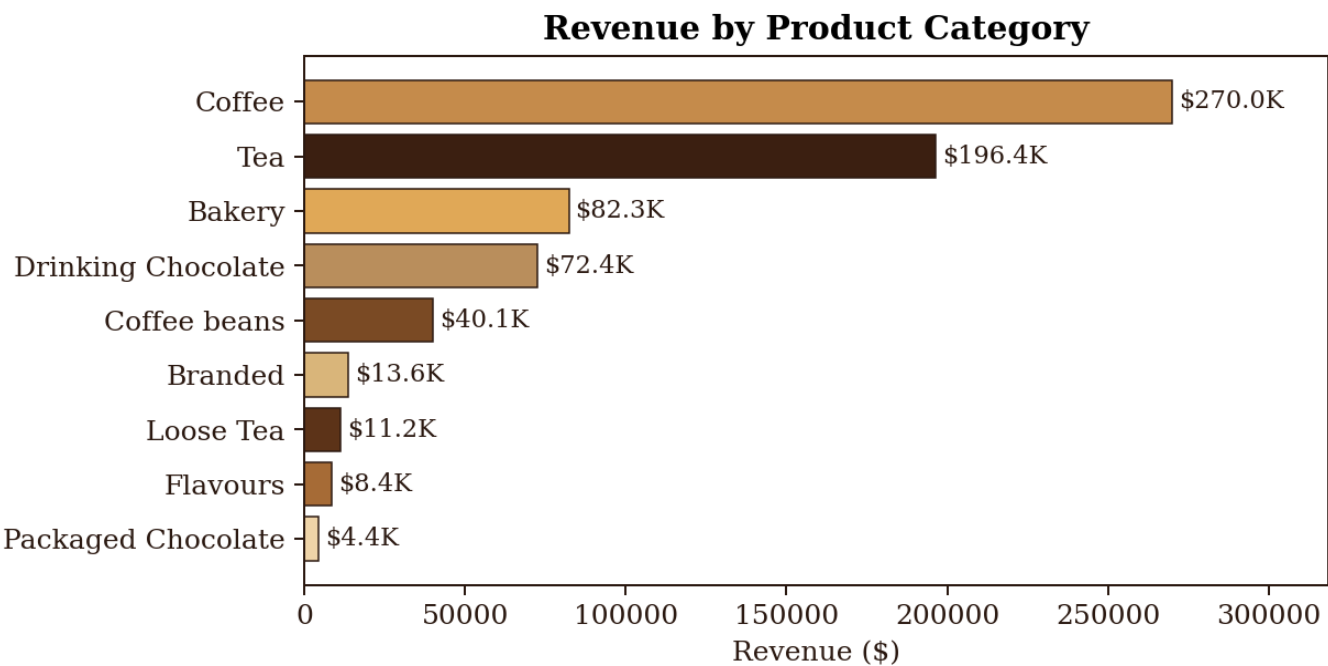
Monthly Revenue Trend (Jan - Jun 2023)



Key Observations

- Revenue grew steadily from **\$81.7K in January** to **\$166.5K in June** — a **+103.8%** lift across the half-year.
- **June** was the strongest month and **February** the weakest, with a noticeable acceleration starting in April that coincides with warmer weather and the introduction of more cold-beverage traffic.
- The trend is monotonically positive after February, suggesting healthy underlying demand rather than seasonality alone — a signal worth confirming with a longer time window.

Revenue by Product Category

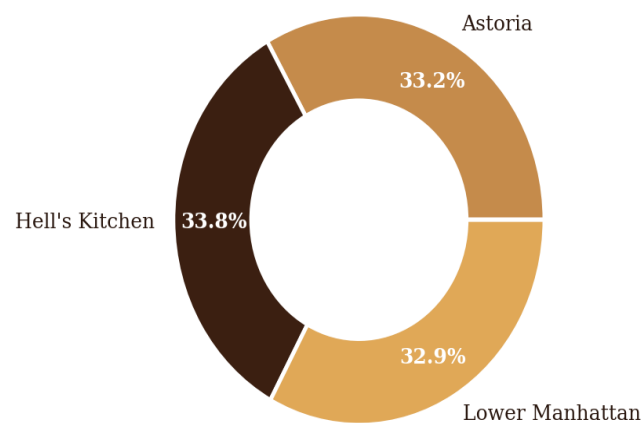


Coffee, Tea and Bakery together account for **\$549K** — roughly **79%** of total revenue. Coffee alone contributes **\$270K** (38.6%), reinforcing it as the anchor category that justifies prime shelf space and supplier focus.

Packaged categories (Coffee Beans, Branded merchandise, Packaged Chocolate) are smaller in dollar terms but carry higher gross margins and warrant secondary placement near the register.

Store Performance

Revenue Share by Store



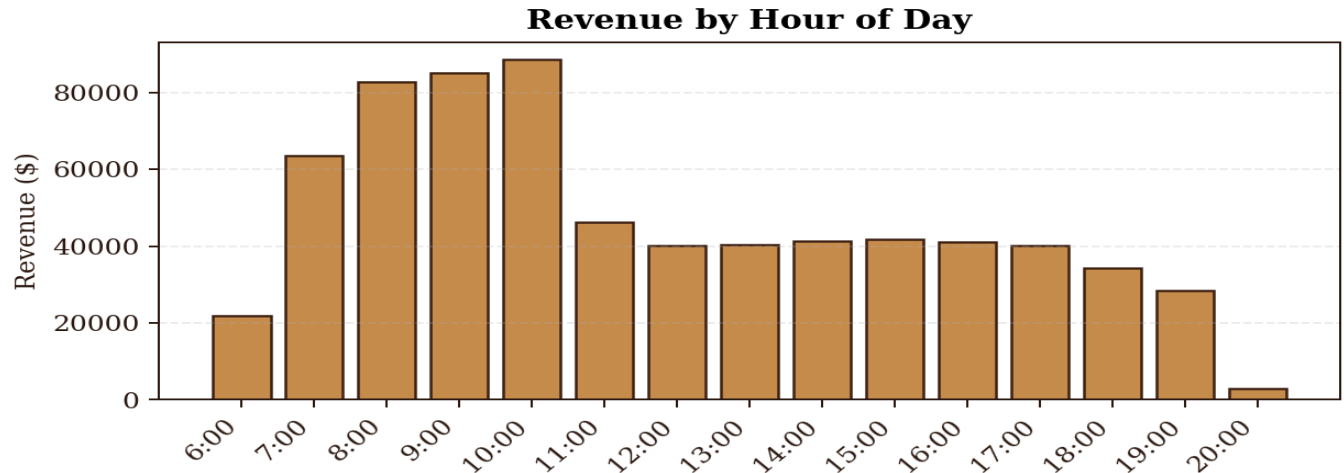
Store	Revenue	Orders	AOV
Astoria	\$232.2K	50,599	\$4.59
Hell's Kitchen	\$236.5K	50,735	\$4.66
Lower Manhattan	\$230.1K	47,782	\$4.81

All three locations perform within roughly **3%** of each other — a reassuring sign that demand is not concentrated in a single neighborhood. **Hell's Kitchen** leads narrowly in revenue, while transaction counts are remarkably consistent across stores ($\approx$ 47K – 51K orders each).

Because order volumes are evenly split, differences in revenue come almost entirely from average ticket size — an actionable lever for upselling, bundling and size-up promotions.

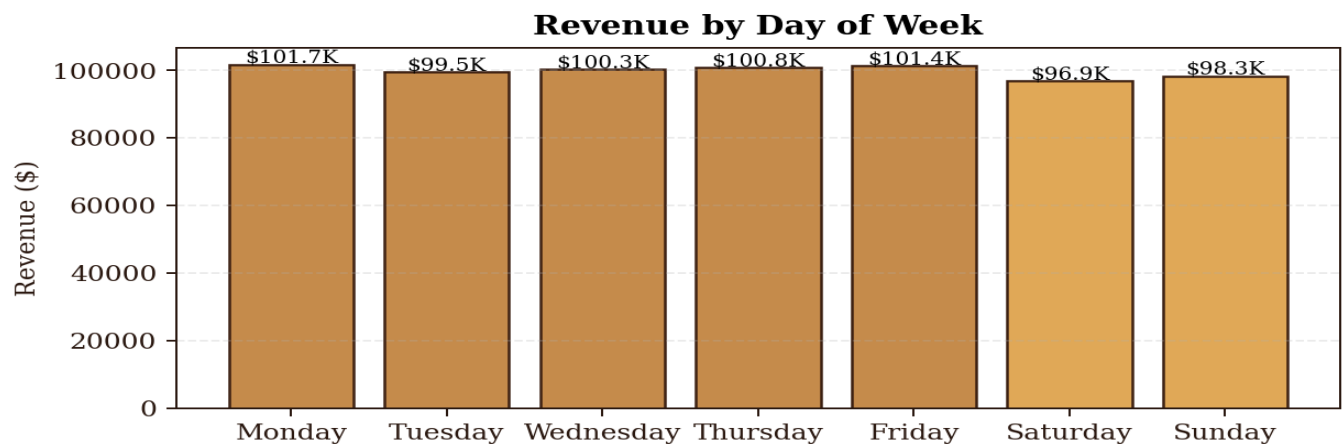
When Customers Buy

Hourly Pattern



A pronounced morning peak runs from **7 AM through 10 AM**, accounting for the tallest revenue bars of the day. Demand softens between 11 AM and 1 PM, rises again into the afternoon (3 – 5 PM) and tapers sharply after 7 PM. Staffing should be heavily front-loaded toward the morning rush.

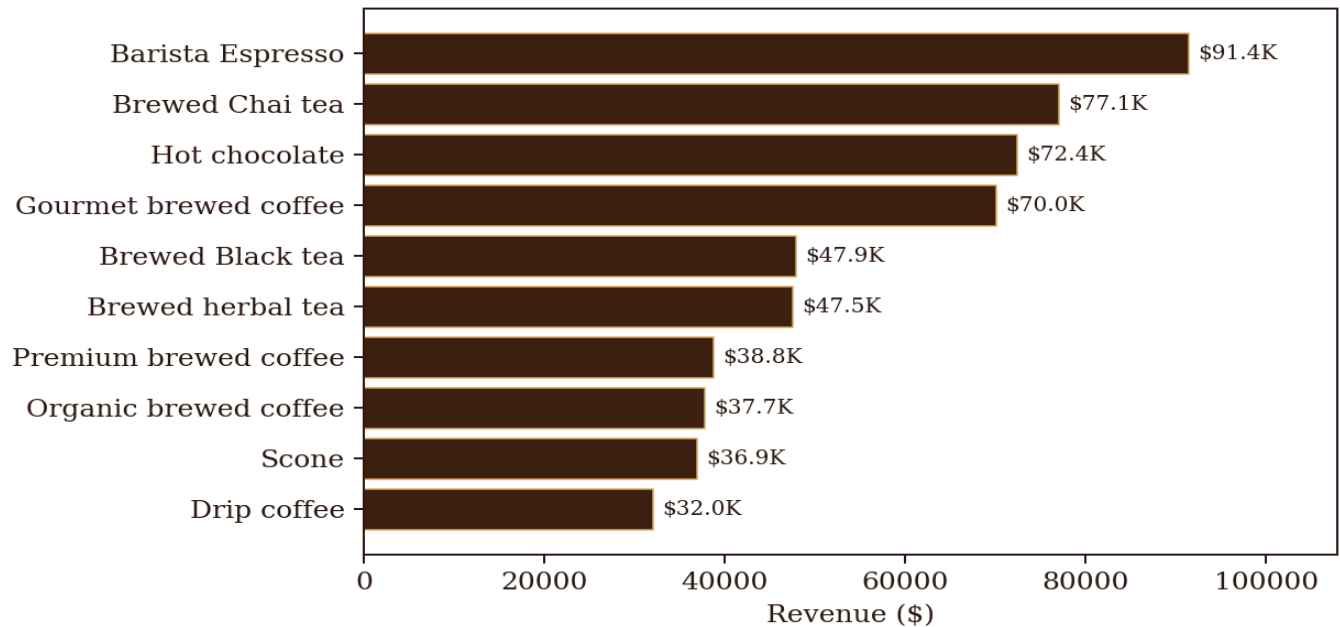
Day of Week



Revenue is impressively flat across the week, with weekdays only marginally ahead of weekends. Monday is the single best day, suggesting a strong commuter base.

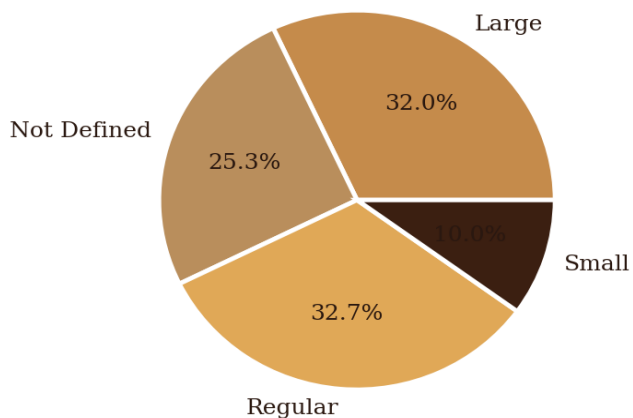
Top Products & Size Mix

Top 10 Product Types by Revenue



Barista Espresso is the single largest revenue contributor, followed by **Brewed Chai Tea** and **Hot Chocolate**. Together the top three product types generate over **\$240K**.

Quantity Sold by Size



Size Mix

Regular and Large dominate quantity sold, with Small the rarest size. The 'Not Defined' bucket reflects items that are not sized (bakery, beans, packaged goods) and represents a quarter of units — useful context when interpreting size-level KPIs.

Key Insights

01

Strong, healthy growth

Revenue climbed ~104% from January to June with no down-month after February — a confident upward trajectory.

02

Coffee is the franchise

Coffee + Tea drive ~67% of all revenue. Any disruption to bean supply or barista throughput is an outsized risk.

03

Mornings are the business

7 – 10 AM is the revenue heart of the day. A 30-minute slip in opening readiness directly costs the largest hour of trade.

04

Stores are interchangeable

Astoria, Hell's Kitchen and Lower Manhattan post nearly identical totals — concept and pricing transfer well across neighborhoods.

05

Average ticket is the lever

\$4.69 AOV. A modest +\$0.50 lift across all 149K transactions would add ~\$75K to the half-year top line.

06

Long tail of products

Beyond the top 5 products, no single SKU exceeds 7% of revenue — pruning slow-movers could simplify operations without hurting sales.

Recommendations & Next Steps

→ **Lock in the morning rush**

Schedule the strongest baristas and pre-stock pastries for the 7–10 AM block. Pilot a "grab-and-go" lane to cut wait times in this window.

→ **Bundle to lift AOV**

Pair top sellers (Barista Espresso, Brewed Chai, Hot Chocolate) with a Scone or pastry add-on at a small discount to nudge AOV above \$5.

→ **Promote in low hours**

Run a 2–3 PM happy-hour offer to smooth the post-lunch dip and convert idle staff capacity into revenue.

→ **Expand high-margin categories**

Coffee Beans and Branded merchandise are tiny today but high-margin. Add an end-cap display and online click-and-collect.

→ **Plan for July+**

Given the strong upward trend, project staffing and inventory using June as the floor — not the average.

→ **Continue measurement**

Re-run this dashboard monthly. Track AOV, morning-hour share, and top-3 product concentration as headline KPIs.

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149,116 transactions · 3 stores · 6 months · \$698,812.33 revenue